Aggression: stability and the selfish machine

***p. 90* … evolutionarily stable strategy …**

I now like to express the essential idea of an ESS in the following more economical way. An ESS is a strategy that does well against copies of itself. The rationale for this is as follows. A successful strategy is one that dominates the population. Therefore it will tend to encounter copies of itself. Therefore it won't stay successful unless it does well against copies of itself. This definition is not so mathematically precise as Maynard Smith's, and it cannot replace his definition because it is actually incomplete. But it does have the virtue of encapsulating, intuitively, the basic ESS idea.

The ESS way of thinking has become more widespread among biologists now than when this chapter was written. Maynard Smith himself has summarized developments up to 1982 in his book *Evolution and the Theory of Games*. Geoffrey Parker, another of the leading contributors to the field, has written a slightly more recent account. Robert Axelrod's *The Evolution of Cooperation* makes use of ESS theory, but I won't discuss it here, since one of my two new chapters, 'Nice guys finish first', is largely devoted to explaining Axelrod's work. My own writings on the subject of ESS theory since the first edition of this book are an article called 'Good Strategy or Evolutionarily Stable Strategy?', and the joint papers on digger wasps discussed in the note to p. 98.

***p. 97* … retaliator emerges as evolutionarily stable.**

This statement, unfortunately, was wrong. There was an error in the original Maynard Smith and Price paper, and I repeated it in this chapter, even exacerbating it by making the rather foolish statement that prober–retaliator is 'almost' an ESS (if a strategy is 'almost' an ESS, then it is not an ESS and will be invaded). Retaliator looks superficially like an ESS because, in a population of retaliators, no other strategy does better. But dove does equally well since, in a population of retaliators, it is indistinguishable in its behaviour from retaliator. Dove therefore can drift into the population. It is what happens next that is the problem. J. S. Gale and the Revd L. J. Eaves did a dynamic computer simulation in which they took a population of model animals through a large number of generations of evolution. They showed that the true ESS in this game is in fact a stable mixture of hawks and bullies. This is not the only error in the early ESS literature that has been exposed by dynamic treatment of this type. Another nice example is an error of my own, discussed in my notes to Chapter 9.

***p. 98* Unfortunately, we know too little at present to assign realistic numbers to the costs and benefits of various outcomes in nature.**

We now have some good field measurements of costs and benefits in nature, which have been plugged into particular ESS models. One of the best examples comes from great golden digger wasps in North America. Digger wasps are not the familiar social wasps of our autumn jam-pots, which are neuter females working for a colony. Each female digger wasp is on her own, and she devotes her life to providing shelter and food for a succession of her larvae. Typically, a female begins by digging a long bore-hole into the earth, at the bottom of which is a hollowed-out chamber. She then sets off to hunt prey (katydids or longhorned grasshoppers in the case of the great golden digger wasp). When she finds one she stings it to paralyse it, and drags it back into her burrow. Having accumulated four or five katydids she lays an egg on the top of the pile and seals up the burrow. The egg hatches into a larva, which feeds on the katydids. The point about the prey being paralysed rather than killed, by the way, is that they don't decay but are eaten alive and are therefore fresh. It was this macabre habit, in the related Ichneumon wasps, that provoked